MOOC Dropout Prediction using Behavioural Sequences and Deep Learning

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Research Domain: Educational Data Mining / Adaptive Learning Systems

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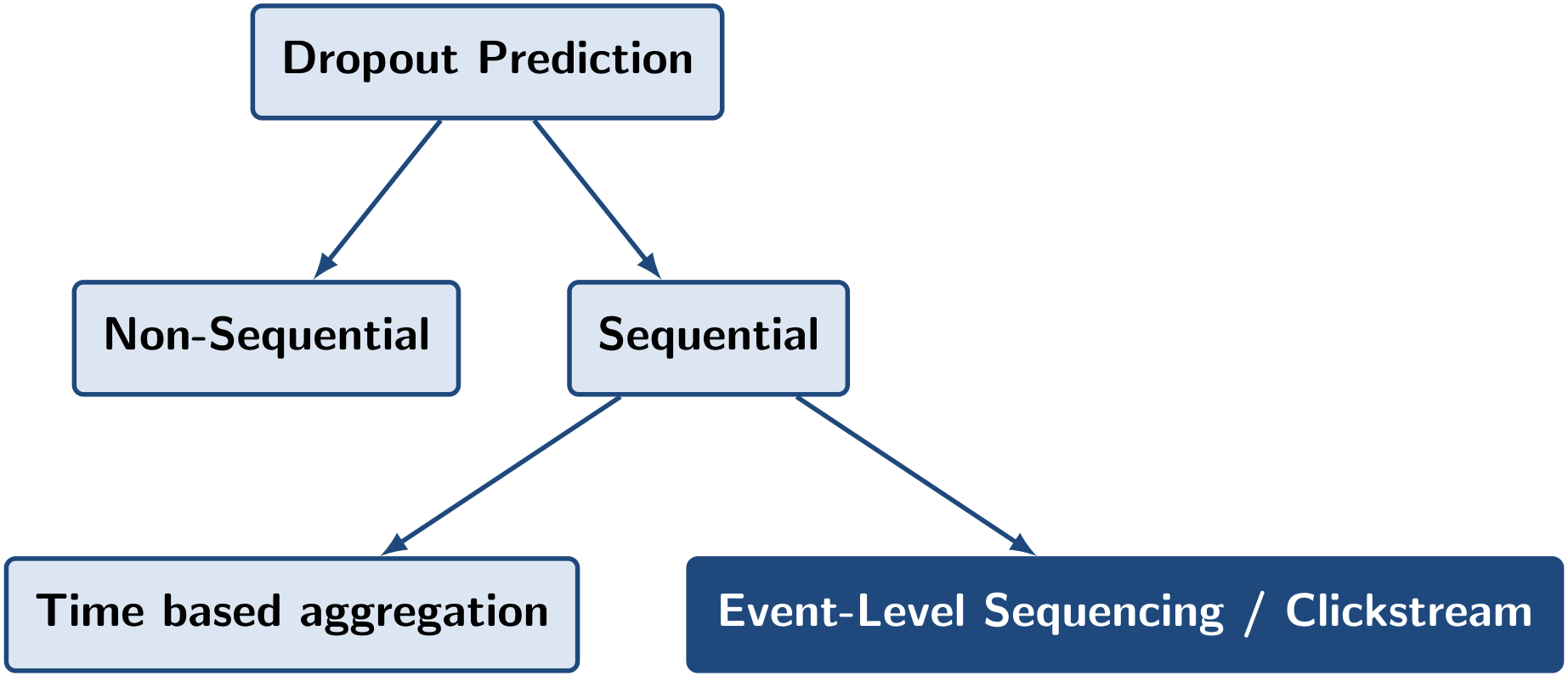
MOOC Dropout Problem

- MOOCs offer free and accessible online courses
- Major issue: very high student dropout rates
- Only about 13% of learners complete the courses
- Nearly 80–85% students drop out before finishing
- Dropout prediction identifies students likely to leave a course early.

Research Area

- Timely dropout prediction by incorporating learning style

Research Classification Hierarchy



Literature Review

- Behavioral Sequences are Predictive [1]
- The "Early Warning" Capability [1]
- Global vs. Local Prediction [1]
- Granularity of Activity [1]
- Privacy-Friendly Analytics [1]

Literature Review

- The hybrid CNN-LSTM approach outperformed standalone models, and achieved an AUC of 84.1% [2]
- Predictive Trends: Their data showed two "danger zones" for student retention
  - The Contact Stage (Weeks 1–2)
  - The Assessment Peak (Weeks 4–5) [2]

Literature Review

Temporal Sequence Modeling in MOOCs

- Modeled dropout prediction as a temporal sequence learning problem [3]
- LSTM significantly outperformed SVM, Logistic Regression, and vanilla RNN models [3]
- Demonstrated the importance of learning long-term behavioral dependencies for MOOC dropout prediction [3]

Attention-Based Sequential Modeling

- Combined GRU with a self-attention mechanism for temporal behavior modeling [4]
- Achieved strong predictive performance with an AUC of 84.79% [4]

Limitations of Existing Models

- Existing dropout prediction models mostly use aggregated temporal features (e.g., weekly video plays, quiz attempts)[1][2][6]
- Sequential patterns in student activity are underutilized
- Learning style information is rarely used to improve dropout prediction[5]

Objectives

- Capture sequential learning patterns in student activity instead of only using aggregated features
- Incorporate learning style information to improve dropout prediction accuracy [5]
- Develop a more effective prediction model that identifies at-risk students earlier
- Improve overall dropout prediction performance for online learning platforms

Dataset Statistics

- KDD cup 2015 Dataset
- Dropout: 95,581 (79.29%)
- Non-dropout: 24,961 (20.71%)
- Students: 79,186
- Enrollments: 120,542
- Courses: 39
- Events per enrollment
  - Avg: 67.67
  - Med: 17
  - Max: 7,697
  - Min: 1

Implementation Overview

Objective

Predict MOOC student dropout using fine-grained behavioral sequences extracted from interaction logs.

Implementation Pipeline

1. Event-level behavioral representation
2. Sequence construction and temporal modeling
3. Multi-stream embedding representation
4. BiLSTM with attention mechanism
5. Temporal aggregation experiments
6. Dropout prediction and evaluation

Core Components

- Source + Event + Category behavioral modeling
- Sliding-window sequence generation
- Attention-enhanced BiLSTM
- Daily and weekly temporal aggregation

Event-Level Behavioral Representation

Key Idea

- Model fine-grained learner interaction sequences instead of aggregated activity counts
- Each interaction event is transformed into a behavioral token

Behavioral Token Construction

$$token = source + "_" + event + "_" + category$$

Examples

- browser_video_video
- browser_problem_problem
- server_access_sequential

Temporal Feature

$$gap = \log(1 + seconds_since_previous_event)$$

Time gaps help capture temporal engagement behavior between interactions.

Sequence Construction and Sliding Window Modeling

Event Ordering

- Events grouped by `enrollment_id`
- Sorted chronologically using timestamps

Sequence Representation

$$(event_token,\ time_gap)$$

Initial Standardization

- Maximum sequence length = 150
- Short sequences padded
- Long sequences truncated

Sliding Window Enhancement

- Window size = 150 events
- Overlap stride = 75 events
- Preserves long-term behavioral information
- Reduces information loss from truncation

Multi-Stream Input Representation

Initial Representation

$$token = source + event + category$$

Entire behavioral token treated as a single embedding unit.

Improved Representation

$$(source,\ event,\ category)$$

Separate Embedding Layers

- Source embedding
- Event embedding
- Category embedding
- Temporal gap feature

Final Event Feature Vector

$$[Source_{emb} \,||\, Event_{emb} \,||\, Category_{emb} \,||\, Gap]$$

This improves semantic representation and reduces vocabulary sparsity.

Attention-Based BiLSTM Architecture

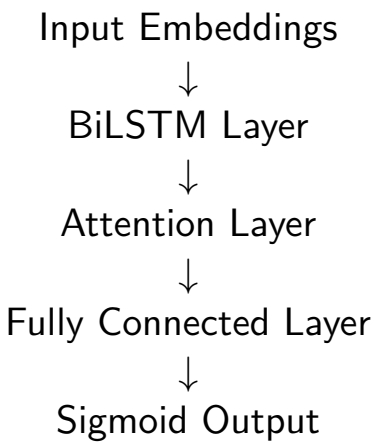
Why BiLSTM?

- Captures temporal behavioral patterns
- Learns both past and future contextual dependencies
- Effective for sequential interaction modeling

Attention Mechanism

- Learns importance weights across sequence timesteps
- Highlights informative learner interactions
- Reduces influence of irrelevant or padded events

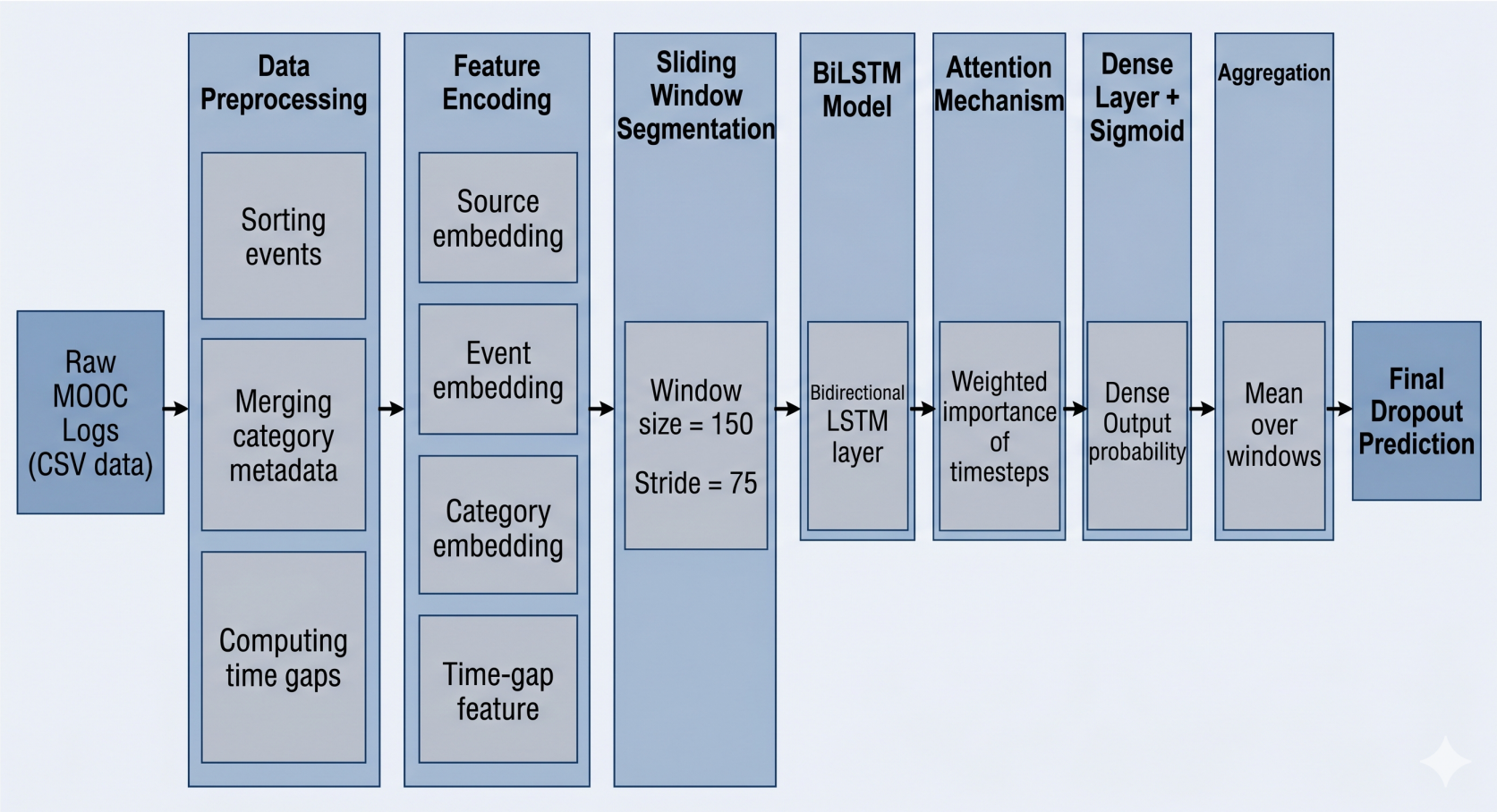
Architecture Flow



Prediction Objective

$$P(\textit{dropout} \mid \textit{behavioral sequence})$$

Model Architecture



Training Configuration

Data Handling

- Sliding-window generation applied to long sequences

Training Parameters

- Loss Function: Binary Cross Entropy
- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 128

Training Objective

Learn behavioral signatures associated with learner disengagement and dropout risk.

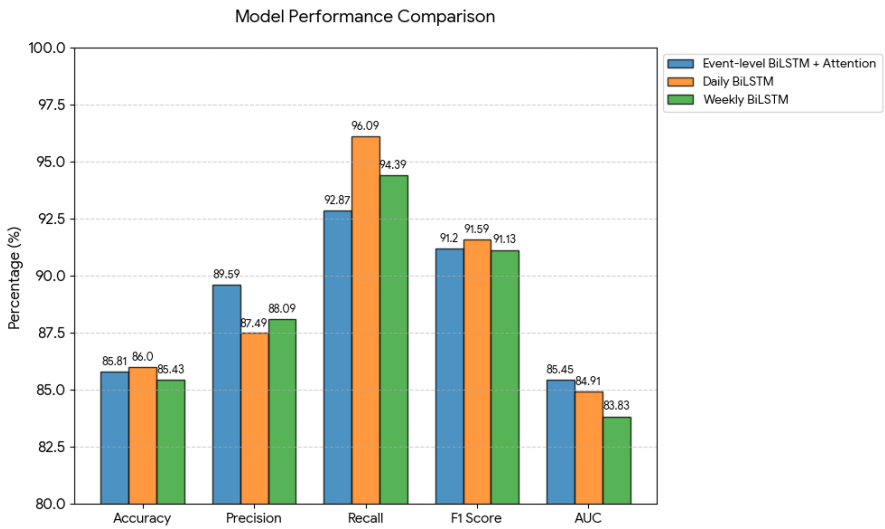
Model Evaluation

Evaluation Metrics

- Accuracy: 0.858
- Precision: 0.895
- Recall: 0.928
- F1 Score: 0.912
- AUC: 0.854

Observations

- Multi-stream embeddings improved feature representation
- Sliding windows preserved long-term learner behavior
- Attention mechanism improved sequence focus
- Temporal aggregation



Evaluation Visualization

Temporal Aggregation Experiments

Motivation

Evaluate how different temporal representations affect dropout prediction performance.

Temporal Modeling Strategies

- Event-level behavioral modeling
- Daily aggregation
- Weekly aggregation

Observations

- Fine-grained event sequences preserve detailed interaction dynamics
- Daily aggregation captures short-term behavioral evolution
- Weekly aggregation reduces sequence complexity
- All evaluated models performed at a similar level across metrics

Temporal Representation Comparison

Performance Comparison of Temporal Modeling Strategies

Model	Acc.	Prec.	Recall	F1	AUC
Event-Level	85.81	89.59	92.87	91.20	85.45
Daily	86.00	87.49	96.09	91.59	84.91
Weekly	85.43	88.09	94.39	91.13	83.83

Key Observations

- All temporal representations achieved comparable overall performance
- Daily aggregation achieved the highest recall and F1-score
- Event-level modeling achieved the highest AUC, indicating stronger discriminative capability
- Weekly aggregation reduced sequence complexity but slightly lowered predictive performance

Conclusion

Temporal aggregation strategies showed only minor variations in performance, indicating that multiple temporal representations can effectively model learner dropout behavior.This shows that event-level sequencing does not significantly outperform daily or weekly aggregation, calling into question the practical necessity of fine-grained sequence modeling for behavioural prediction.

Early Detection using First 2 Weeks of Data

Objective

Evaluate dropout prediction performance using only the first two weeks of learner interaction data.

Performance Comparison (2 Weeks)

Model	Acc.	Prec.	Recall	F1	AUC
Event-Level	83.99	87.30	93.36	90.23	83.04
Daily	75.62	92.23	75.58	83.08	80.59
Weekly	34.57	96.49	18.03	30.39	79.74

Key Observations

- Event-level model achieved strong early prediction performance
- High recall (93%) and F1-score (90%) indicate effective dropout detection
- Daily aggregation showed moderate performance with high precision but lower recall
- Weekly aggregation failed to capture early behavioral signals effectively

Takeaway

Fine-grained event-level modeling significantly improves early-stage dropout detection.

Early Detection using First Week of Data

Objective

Evaluate model performance under extreme data limitation using only the first week of learner activity.

Performance Comparison (1 Week)

Model	Acc.	Prec.	Recall	F1	AUC
Event-Level	82.19	83.91	95.90	89.51	80.08
Daily	35.90	72.09	31.10	43.45	44.78

Key Observations

- Event-level model remained highly effective even with limited data
- Recall of 95.9% indicates strong early dropout identification capability
- Daily aggregation performance degraded significantly
- Performance gap widened substantially during early-stage prediction

Note

Weekly aggregation was excluded since aggregation is not meaningful at one-week granularity.

Early Detection Insights

Impact of Temporal Granularity

Event-Level Modeling

- Captures fine-grained learner interactions and transitions
- Retains predictive power even with limited behavioral history

Daily Aggregation

- Relies on aggregated behavioral statistics
- Requires longer observation periods for stable prediction

Weekly Aggregation

- Over-compresses temporal behavioral information
- Fails to capture subtle early-stage dropout signals

Key Insight

Early dropout behavior is subtle, short-term, and highly sequential in nature.

Conclusion

Temporal resolution is critical for early intervention systems, with event-level behavioral modeling enabling reliable prediction even within the first week.

Implementation Summary

Key Contributions

- Event-level behavioral sequence modeling
- Multi-stream embedding representation
- Sliding-window sequence generation
- Attention-enhanced BiLSTM architecture
- Temporal aggregation experiments

Outcome

The proposed implementation captured richer behavioral patterns while improving sequence representation and temporal modeling for MOOC dropout prediction.

Limitation

Sliding-window modeling improves sequence coverage but may still fail to capture the complete learner behavioral trajectory across the entire course duration.

Future Improvements

- **Addressing Current Drawbacks** The current sliding-window approach improves long-sequence handling; however, it may still fail to capture the complete behavioral context of a learner across the entire course duration.
- **Advanced Sequence Models**

  Explore architectures such as:

  - Transformer-based sequence models

  These models can better capture long-range dependencies in student behavior.
- **Model Interpretability**

  Apply interpretability techniques to understand which behavioral patterns contribute most to dropout predictions.
- **Learning Style Integration**

  Incorporate results from a learning style detector and analyze whether student learning styles correlate with dropout behavior.[5]

Conclusion

Overall Findings

- All temporal representations achieved comparable overall performance
- Event-level modeling showed slightly better discriminative capability based on AUC
- Daily aggregation achieved higher recall in the full-data setting

Early Prediction Insight

- Event-level modeling retained strong predictive performance even with limited data
- Achieved high recall within the first 1–2 weeks of learner activity
- Aggregated temporal models required longer observation periods for stable prediction

Key Takeaway

Temporal granularity does not significantly affect overall prediction performance, but becomes highly important for early-stage dropout detection.

Final Conclusion

Fine-grained behavioral modeling may not always be necessary for full-course prediction, but is essential for timely intervention and early warning systems.

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